Efficacy and safety of finerenone in patients with chronic kidney disease: an individual participant data pooled analysis (INFINITY)

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Summary

Background Overactivation of the mineralocorticoid receptor is a common pathway for chronic kidney disease (CKD) progression across multiple disease aetiologies. Finerenone, a non-steroidal mineralocorticoid receptor antagonist, has shown kidney and cardiovascular benefits in CKD due to type 2 diabetes, but its efficacy and safety across disease aetiologies, and levels of glycaemia, estimated glomerular filtration rate (eGFR), and albuminuria have not been evaluated. The aim of this study was to evaluate the efficacy and safety of finerenone across the spectrum of CKD.

Methods We conducted an individual participant data meta-analysis of three randomised, double-blind, placebo-controlled trials of finerenone in patients with CKD: FIDELIO-DKD (NCT02540993; Sept 17, 2015, to April 14, 2020), FIGARO-DKD (NCT02545049; Sept 17, 2015, to Feb 2, 2021), and FIND-CKD (NCT05047263; Sept 21, 2021, to Feb 2, 2026). We used Cox regression models to evaluate relative effects on kidney and cardiovascular outcomes. The main kidney outcome was kidney failure or sustained 57% or more decline in eGFR; the main cardiovascular outcome was hospitalisation for heart failure or cardiovascular death. This study was registered with PROSPERO, CRD420251269149.

Findings Across the three trials enrolling 14 574 participants, the mean age was 63·7 years (SD 10·6), 4467 (30·7%) were female, 10 107 (69·3%) were male, mean eGFR was 56·4 mL/min per 1·73 m² (SD 21·4), and median urinary albumin-to-creatinine ratio was 567·4 mg/g (IQR 233·6–1164·7). Finerenone reduced the risk of the composite kidney outcome by 24% versus placebo (22·3 *vs* 28·8 events per 1000 patient-years; hazard ratio 0·76 [95% CI 0·68–0·86]) and kidney failure alone (0·85 [0·74–0·99]). Finerenone reduced the risk of the composite cardiovascular outcome versus placebo (19·1 *vs* 23·9 events per 1000 patient-years; 0·80 [0·70–0·91]), including heart failure hospitalisation (0·78 [0·66–0·92]) and cardiovascular death (0·82 [0·67–0·999]). Finerenone also reduced the risk of all-cause death (0·88 [0·79–0·99]). Treatment effects on the composite kidney outcome were consistent irrespective of glycaemic status, CKD aetiology, baseline eGFR, albuminuria, and use of sodium–glucose co-transporter-2 inhibitors. Hyperkalaemia occurred more frequently with finerenone than with placebo, but the absolute incidence of hyperkalaemia leading to hospitalisation was low.

Interpretation In the studied populations with CKD, finerenone reduced the risk of CKD progression, including kidney failure alone, and reduced heart failure hospitalisation, cardiovascular death, and all-cause death. These findings support finerenone as a foundational therapy for CKD across a broad range of disease aetiologies and levels of glycaemia, eGFR, and albuminuria.

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Introduction

Overactivation of the mineralocorticoid receptor is a key driver of inflammation and fibrosis in the kidneys, heart, and vasculature, representing a common mechanism underlying disease progression in chronic kidney disease (CKD), and related conditions, including hypertension and heart failure.[1] Finerenone is a selective, non-steroidal mineralocorticoid receptor antagonist that blocks mineralocorticoid receptor-mediated sodium retention and, in experimental models, leads to structural improvements in cardiac and kidney tissue and reductions in multiple biomarkers implicated in inflammation and fibrosis.[2,3]

In the FIDELIO-DKD and FIGARO-DKD trials, finerenone reduced the risk of kidney and cardiovascular outcomes in 12 990 individuals with CKD associated with

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Research in context

Evidence before this study

In a systematic review conducted in 2024, we searched PubMed and MEDLINE from database inception to July 24, 2024, for phase 3, randomised, double-blind, placebo-controlled trials evaluating the efficacy and safety of finerenone, a nonsteroidal mineralocorticoid receptor antagonist, using the terms "finerenone" and "randomized controlled trial". We applied no language restrictions to this search. The review identified two phase 3 trials in patients with chronic kidney disease (CKD) and type 2 diabetes: FIDELIO-DKD and FIGARO-DKD. Together, these trials enrolled 12 990 participants with a median follow-up of 3 years. The trials showed that finerenone reduced the risk of CKD progression and cardiovascular events, findings that supported regulatory approval and subsequent recommendations in clinical practice guidelines for this population. However, most cases of CKD worldwide are not attributable to diabetes, and at the time of the systematic review there was no robust evidence regarding the efficacy or safety of finerenone in patients with CKD without diabetes. Since then, the FINEARTS-HF trial showed that finerenone reduced heart failure events in 6001 individuals with heart failure with mildly reduced or preserved ejection fraction. Although this trial included individuals without diabetes, it was not designed to assess kidney outcomes. Consequently, guideline recommendations for finerenone in CKD have remained restricted to patients with diabetes.

Added value of this study

More recently, the FIND-CKD trial showed that finerenone reduced the rate of kidney function loss in patients with CKD who did not have diabetes, extending the evidence for finerenone beyond CKD and type 2 diabetes, but without providing a comprehensive evaluation of effects on kidney and cardiovascular outcomes across the full spectrum of CKD, or reliable estimates on outcomes such as kidney failure and mortality. To address this limitation, we updated the previous systematic review and conducted an individual participant data meta-analysis of the FIDELIO-DKD, FIGARO-DKD, and FIND-CKD trials. In 14 574 individuals followed up for a median of 3·1 years, finerenone reduced the risk of a composite kidney outcome comprising kidney failure or sustained 57% or more decline in estimated glomerular filtration rate (eGFR; hazard ratio 0·76 [95% CI 0·68–0·86]), with clear reduction in kidney failure alone (0·85 [0·74–0·99]). Finerenone reduced the risk of heart failure hospitalisation or cardiovascular death (0·80 [0·70–0·91]), driven by both heart failure hospitalisation (0·78 [0·66–0·92]) and cardiovascular death (0·82 [0·67–0·999]). In addition, finerenone also reduced the risk of all-cause death (0·88 [0·79–0·99]). Effects on the composite kidney and cardiovascular outcomes were broadly consistent across a range of prespecified subgroups, including glycaemic status, cause of kidney disease, eGFR, albuminuria, and use of sodium–glucose co-transporter-2 inhibitors. Although finerenone increased the incidence of hyperkalaemia compared with placebo (14·3% *vs* 7·6%), events leading to hospitalisation were uncommon (0·9% *vs* 0·2%) with no deaths attributable to hyperkalaemia. Although the absolute risk of cardiovascular events and mortality differed by diabetes status, absolute estimates showed that the net kidney, cardiovascular, and mortality benefits of finerenone outweighed adverse effects, irrespective of diabetes status.

Implications of all the available evidence

The totality of the randomised evidence supports the role of finerenone as a foundational therapy to improve kidney and cardiovascular outcomes in patients with CKD across a broad range of disease aetiologies, spectrum of glycaemia, eGFR, and albuminuria.

type 2 diabetes.[4–6] These findings supported regulatory approval of finerenone in this population and have informed major clinical practice guideline recommendations on its use in CKD associated with type 2 diabetes.[7–10] More recently, the FIND-CKD trial evaluated the effects of finerenone in 1584 patients with non-diabetic CKD spanning a broad range of underlying aetiologies.[11] Because the primary objective of FIDELIO-DKD was to determine the effect of finerenone on clinical kidney outcomes, the trial preferentially enrolled participants with higher levels of albuminuria.[5] In contrast, FIGARO-DKD enrolled participants with lower albuminuria and a wider range of kidney function to evaluate the effect of finerenone on a primary cardiovascular outcome.[6] Finally, the FIND-CKD trial used annualised decline (slope) in estimated glomerular filtration rate (eGFR), a validated surrogate for CKD progression, as its primary endpoint, reflecting its aim to assess the effects of finerenone on CKD progression in an adjacent population (non-diabetic CKD) after safety had been shown in the preceding FIDELIO-DKD and FIGARO-DKD trials.[5,6,11–13]

The distinct but overlapping populations enrolled across these three trials provide a unique opportunity to comprehensively evaluate the relative and absolute effects of finerenone on kidney, cardiovascular, and safety outcomes across the spectrum of CKD. These populations include patients differing in glycaemic status, kidney function, and underlying kidney disease aetiology, and therefore at differing risks of kidney failure, cardiovascular events, and hyperkalaemia. In addition, although the individual trials suggested potential benefits for kidney failure and mortality outcomes, none were designed to definitively assess these endpoints. We therefore conducted INFINITY, a prespecified pooled individual participant data analysis of the FIDELIO-DKD, FIGARO-DKD, and FIND-CKD trials, to generate reliable estimates of the efficacy and safety of finerenone in patients across the full spectrum of CKD.

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Methods

Search strategy and selection criteria

We pooled individual participant-level data from phase 3, multicentre, double-blind trials involving participants with CKD and type 2 diabetes (FIDELIO-DKD; NCT02540993 and FIGARO-DKD; NCT02545049) and non-diabetic CKD (FIND-CKD; NCT05047263). The study designs of the three trials and details of a previous pooled analysis of the FIDELIO-DKD and FIGARO-DKD trials (FIDELITY) have been published previously,[4,11,14,15] and are summarised in the appendix (pp 17–18).

In addition to the three prespecified trials, we sought to identify any other eligible phase 3 trials for potential inclusion. We conducted a systematic search of PubMed and Embase from database inception to March 31, 2026, without language restriction for phase 3, randomised, double-blind, placebo-controlled trials evaluating the efficacy and safety of finerenone in primary adult CKD populations with at least 500 participants per treatment group and at least 6 months' follow-up. Trials enrolling mixed populations with and without CKD were excluded. Earlier-phase studies, open-label trials, and trials without available individual participant-level data were also excluded. The full search strategy is in the appendix (p 19). The protocol for this individual participant data pooled analysis was prospectively registered in PROSPERO (CRD 420251269149) before the completion of the FIND-CKD trial.

FIDELIO-DKD and FIGARO-DKD

Briefly, the FIDELIO-DKD (Sept 17, 2015, to April 14, 2020) and FIGARO-DKD (Sept 17, 2015, to Feb 2, 2021) trials enrolled adults (aged ≥18 years) with CKD and type 2 diabetes treated with a maximum tolerated dose of a renin–angiotensin system inhibitor for at least 4 weeks before screening.[5,6] In FIDELIO-DKD, participants had an eGFR of ≥25 to <60 mL/min per 1·73 m² with a urinary albumin-to-creatinine ratio (UACR) of ≥30 to <300 mg/g and a history of diabetic retinopathy, or eGFR ≥25 to <75 mL/min per 1·73 m² with a UACR of ≥300 to ≤5000 mg/g.[5] In FIGARO-DKD, participants had an eGFR of ≥25 to <90 mL/min per 1·73 m² with a UACR of ≥30 to <300 mg/g, or eGFR ≥60 mL/min per 1·73 m² with a UACR of ≥300 to ≤5000 mg/g.[6] Serum potassium was required to be ≤4·8 mmol/L at both the run-in and screening visits in both trials.[5,6] A diagnosis of non-diabetic CKD was an exclusion criterion in both trials.[5,6]

FIND-CKD

In the FIND-CKD trial (Sept 21, 2021, to Feb 2, 2026), individuals were eligible if they had an eGFR of ≥25 to <60 mL/min per 1·73 m² with a UACR of ≥200 to <500 mg/g or an eGFR of ≥25 to <90 mL/min per 1·73 m² with a UACR ≥500 to ≤3500 mg/g. All participants were required to be receiving a stable or maximum tolerated dose of a renin–angiotensin system inhibitor for at least 4 weeks with a serum potassium level ≤4·8 mmol/L. Key exclusion criteria included a diagnosis of type 1 diabetes, type 2 diabetes, or glycated haemoglobin ≥6·5%. Participants with immunological kidney diseases requiring immunosuppression and participants with cystic kidney diseases were also excluded.[13] Patients with heart failure with reduced ejection fraction were excluded from all three trials because they had an existing guideline-based indication for a steroidal mineralocorticoid receptor antagonist.[5,6,11,13]

Key trial procedures

In all three trials, eligible participants were randomised 1:1 to finerenone (oral tablet) or placebo. Initial dosing was determined by eGFR levels at screening; 10 mg once daily for patients with an eGFR of 25 to <60 mL/min per 1·73 m² and 20 mg once daily for patients with an eGFR ≥60 mL/min per 1·73 m². From month 1 onward, participants assigned to finerenone 10 mg could be up-titrated to the target dose of 20 mg, provided their serum potassium level was ≤4·8 mmol/L and eGFR was stable; down-titration to 10 mg once daily was permitted at any time after treatment initiation. Study drug was withheld if potassium concentrations were more than 5·5 mmol/L and restarted at 10 mg when they decreased to 5·0 mmol/L or less.[5,6,13]

Outcomes

The main composite kidney outcome in the present study was a sustained 57% or more decline in eGFR relative to baseline (corresponding approximately to a doubling of serum creatinine) or kidney failure. Kidney failure was defined as sustained eGFR of less than 15 mL/min per 1·73 m² or initiation of chronic dialysis or kidney transplantation. In addition, we evaluated several other kidney and composite cardiovascular and kidney outcomes, including (1) kidney failure alone; (2) the composite kidney outcome inclusive of cardiovascular death; and (3) the composite kidney outcome or hospitalisation for heart failure or cardiovascular death (ie, the main kidney and cardiovascular outcome). In additional analyses, sustained 40% or more and 50% or more declines in eGFR relative to baseline were substituted for a ≥57% or more decline as a component of the composite kidney outcome. An outcome comprising chronic dialysis or kidney transplantation was assessed post-hoc. eGFR was calculated according to the CKD-EPI 2009 creatinine equation with adjustment for race. The effect of finerenone on the main composite kidney outcome was evaluated across prespecified subgroups defined at baseline, including age, sex, region, systolic blood pressure, BMI, HbA_{1c}, history of cardiovascular disease, heart failure, eGFR, UACR, Kidney Disease Improving Global Outcomes (KDIGO) risk category, and SGLT2 inhibitor use.

The main cardiovascular outcome in the present study was a composite of hospitalisation for heart failure or cardiovascular death. Cardiovascular death in this

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See **Online** for appendix

analysis excluded deaths classified as undetermined. Other cardiovascular, hospitalisation, and mortality outcomes assessed included new-onset atrial fibrillation; non-fatal myocardial infarction, non-fatal stroke, hospitalisation for heart failure, or cardiovascular death; hospitalisation for heart failure; all-cause hospitalisation; cardiovascular death; and all-cause death. Cardiovascular and mortality outcomes were adjudicated in FIDELIO-DKD and FIGARO-DKD but not FIND-CKD, where these events were investigator-reported and prospectively recorded in electronic case report forms specifically designed to capture these outcomes. Definitions for these outcomes and approach to harmonisation across trials are in the appendix (p 20).

In the present study, a range of prespecified safety outcomes were assessed, including serious adverse events, adverse events leading to treatment discontinuation, acute kidney injury, and hypotension. Potassium-related adverse events included investigator-reported hyperkalaemia (identified by MedDRA terms “hyperkalaemia” or “blood potassium increased”), hyperkalaemia leading to treatment discontinuation, hyperkalaemia resulting in hospitalisation, hyperkalaemia resulting in death, and central laboratory-measured serum potassium values of more than 5·5 mmol/L, more than 6·0 mmol/L, and less than 3·5 mmol/L.

	Finerenone (n=7291)	Placebo (n=7283)	Overall (n=14 574)
Age, years	63·6 (10·5)	63·7 (10·7)	63·7 (10·6)
Sex*			
Female	2310 (31·7%)	2157 (29·6%)	4467 (30·7%)
Male	4981 (68·3%)	5126 (70·4%)	10 107 (69·3%)
Race*			
White	4757 (65·2%)	4760 (65·4%)	9517 (65·3%)
Black	273 (3·7%)	284 (3·9%)	557 (3·8%)
Asian	1860 (25·5%)	1868 (25·6%)	3728 (25·6%)
Other†	401 (5·5%)	371 (5·1%)	772 (5·3%)
Region			
Europe, Australia, and Israel	3527 (48·4%)	3550 (48·7%)	7077 (48·6%)
Asia	1750 (24·0%)	1734 (23·8%)	3484 (23·9%)
North America	1096 (15·0%)	1085 (14·9%)	2181 (15·0%)
Latin America	753 (10·3%)	754 (10·4%)	1507 (10·3%)
Others	165 (2·3%)	160 (2·2%)	325 (2·2%)
eGFR, mL/min per 1·73 m²	56·3 (21·3)	56·4 (21·5)	56·4 (21·4)
eGFR category, mL/min per 1·73 m²			
<45	2617 (35·9%)	2613 (35·9%)	5230 (35·9%)
45 to <60	1918 (26·3%)	1931 (26·5%)	3849 (26·4%)
≥60	2755 (37·8%)	2737 (37·6%)	5492 (37·7%)
UACR, mg/g	566·6 (231·8–1155·5)	568·3 (235·2–1173·2)	567·4 (233·6–1164·7)
UACR category, mg/g			
<300	2218 (30·4%)	2155 (29·6%)	4373 (30·0%)
300 to ≤1000	2884 (39·6%)	2917 (40·1%)	5801 (39·8%)
>1000	2187 (30·0%)	2208 (30·3%)	4395 (30·2%)
BMI, kg/m²	30·9 (6·1)	30·9 (6·0)	30·9 (6·1)
Mean systolic blood pressure, mm Hg	136·0 (14·3)	135·9 (14·4)	136·0 (14·4)
Mean HbA_{1c}, %	7·5 (1·5)	7·5 (1·5)	7·5 (1·5)
Potassium, mmol/L	4·4 (0·4)	4·4 (0·4)	4·4 (0·4)
History of cardiovascular disease			
Atherosclerotic cardiovascular disease	3063 (42·0%)	3060 (42·0%)	6123 (42·0%)
Atrial fibrillation or atrial flutter	594 (8·1%)	571 (7·8%)	1165 (8·0%)
Heart failure	501 (6·9%)	541 (7·4%)	1042 (7·1%)
Concomitant medications			
Angiotensin-converting enzyme inhibitors	2753 (37·8%)	2758 (37·9%)	5511 (37·8%)
Angiotensin receptor blockers	4529 (62·1%)	4520 (62·1%)	9049 (62·1%)
SGLT2 inhibitors	570 (7·8%)	572 (7·9%)	1142 (7·8%)
Statins	5068 (69·5%)	5193 (71·3%)	10 261 (70·4%)

Data are reported as n (%), mean (SD), or median (IQR). eGFR=estimated glomerular filtration rate. FSGS=focal segmental glomerulosclerosis. HbA_{1c}=glycated haemoglobin. SGLT2=sodium–glucose co-transporter-2. UACR=urine albumin-to-creatinine ratio. *Sex and race were self-reported by the participants. Participants reporting more than one race were categorised as multiple race. †American Indian or Alaska native, Hawaiian or other Pacific Islander, or not reported, or multiple.

Table 1: **Characteristics of participants at baseline**

Data analysis

The full analysis set comprised all randomly assigned patients (except 72 patients from FIDELIO-DKD and 109 from FIGARO-DKD with critical Good Clinical Practice [GCP] violations, who were prospectively excluded from all analyses). Safety analyses were performed in the safety analysis set, defined as all randomly assigned patients without critical GCP violations who took one or more dose of the study drug.

Baseline characteristics of participants were reported as numbers, proportions, and mean (SD) for normally distributed variables, and median (IQR) values for skewed variables.

Key clinical outcomes were graphically displayed using Aalen–Johansen methods accounting for mortality as a competing risk. All clinical efficacy outcomes were analysed using stratified Cox proportional hazards models fitted using the stratification factors UACR at screening (≤1000 mg/g and >1000 mg/g), SGLT2 inhibitor use at baseline (yes or no), and trial. Heterogeneity among trials was assessed based on a Cox proportional hazards model and including the covariates treatment, trial, and the corresponding interaction. Subgroup analyses were conducted based on stratified Cox models fitted across subgroup categories, including covariates treatment, subgroup, and the respective interaction. Heterogeneity between subgroups was assessed via the interaction p value. Adverse event-based analyses included incidence proportions and exposure-adjusted incidences per 1000 patient-years by treatment allocation.

Because the absolute risks of cardiovascular events, mortality, and hyperkalaemia differ in patients with CKD according to diabetic versus non-diabetic aetiology, we estimated the absolute effects of finerenone on key efficacy and safety outcomes both overall and stratified by diabetes status. Absolute event rates per 1000 patient-years were modelled using Poisson regression. For outcomes in which no heterogeneity by diabetes status was observed, the overall relative treatment effect was applied to these baseline event rates in the placebo group. When evidence of interaction by diabetes status was present, diabetes-specific relative treatment effects were applied to the corresponding placebo event rates within each subgroup. The primary safety outcomes assessed were hyperkalaemia and acute kidney injury, defined as events leading to hospitalisation, to focus on clinically meaningful events. Absolute effects were presented as the number of events averted or caused per 1000 patient-years of treatment over 3 years, aligning with the median duration of follow-up in INFINITY. Numbers needed to treat and harm for key efficacy and safety outcomes, respectively, were calculated to complement absolute effect estimates.

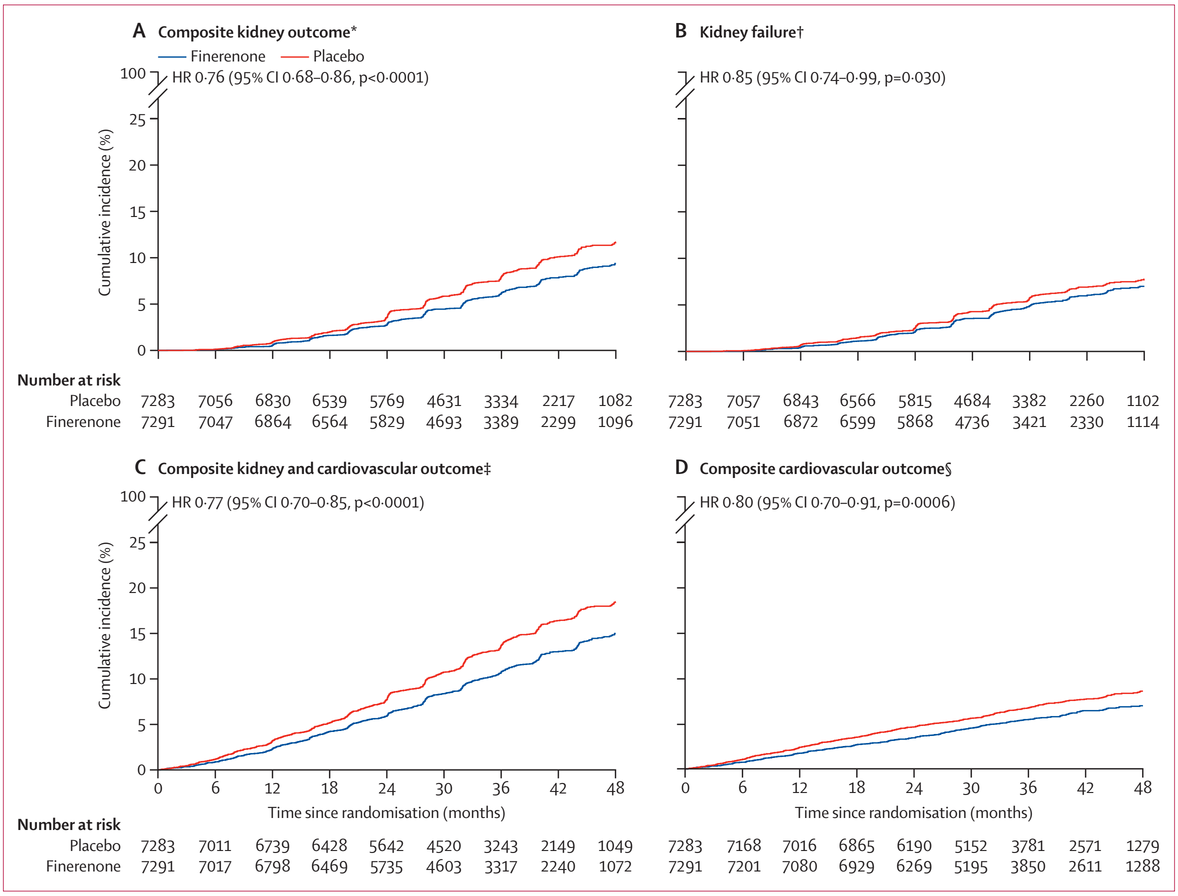
All statistical analyses were prespecified but exploratory, rather than hypothesis-confirming; therefore, p values were reported with no adjustment for multiplicity. Statistical analyses were performed either by authors at University Medical Center Groningen using R software version 4.4 or by Bayer using SAS version 9.4.

Role of the funding source

The FIDELIO-DKD, FIGARO-DKD, and FIND-CKD trials, and the INFINITY pooled analysis were funded by Bayer. All data in the present study were analysed by authors at University Medical Center Groningen and independently verified by Bayer. Bayer, in cooperation with the academic steering committees, designed and oversaw the conduct of the trials. Bayer conducted site monitoring, study-level data collection, and dataset creation. Bayer reviewed the publication and could provide comments but data interpretation was made only by the authors. Bayer-affiliated authors contributed to the manuscript in accordance with authorship criteria.

Results

Our systematic review yielded three eligible trials, FIDELIO-DKD, FIGARO-DKD, and FIND-CKD, with no additional studies meeting inclusion criteria (appendix p 25). All three trials were assessed as having a low risk of bias (appendix p 26). The overall INFINITY pooled population comprised 14 574 participants: the mean age



Number at risk (A)									
Placebo	7283	7056	6830	6539	5769	4631	3334	2217	1082
Finerenone	7291	7047	6864	6564	5829	4693	3389	2299	1096

Number at risk (B)									
Placebo	7283	7057	6843	6566	5815	4684	3382	2260	1102
Finerenone	7291	7051	6872	6599	5868	4736	3421	2330	1114

Number at risk (C)									
Placebo	7283	7011	6739	6428	5642	4520	3243	2149	1049
Finerenone	7291	7017	6798	6469	5735	4603	3317	2240	1072

Number at risk (D)									
Placebo	7283	7168	7016	6865	6190	5152	3781	2571	1279
Finerenone	7291	7201	7080	6929	6269	5195	3850	2611	1288

***Figure 1:* Cumulative incidence functions for key efficacy outcomes**

Cumulative incidence curves of efficacy outcomes estimated using Aalen–Johansen methods accounting for mortality as a competing risk. Outcomes were assessed in time-to-event analyses. eGFR=estimated glomerular filtration rate. HR=hazard ratio. *The composite kidney outcome was a 57% or more decline in eGFR or kidney failure. †Kidney failure was defined as sustained eGFR less than 15 mL/min per 1·73 m² or initiation of chronic dialysis or kidney transplantation. ‡Comprises the composite kidney outcome, or heart failure hospitalisation or cardiovascular death. §Heart failure hospitalisation or cardiovascular death.

was 63·7 years (SD 10·6), 4467 (30·7%) were female, 10 107 (69·3%) were male, the mean eGFR was 56·4 mL/min per 1·73 m² (SD 21·4), and the median UACR was 567·4 mg/g (IQR 233·6–1164·7; table 1). 6123 (42·0%) participants had a history of atherosclerotic cardiovascular disease and 1042 (7·1%) had a recorded diagnosis of heart failure; 1142 (7·8%) were receiving SGLT2 inhibitors at baseline. Baseline characteristics by individual trial and distribution of eGFR and UACR by trial are in the appendix (pp 21, 27).

Over a median follow-up of 3·1 years (IQR 2·4–3·8), there were 1049 (7·2%) participants who had the main

	Events or participants		Event rate per 1000 patient-years		HR (95% CI)	p value
	Finerenone (n=7291)	Placebo (n=7283)	Finerenone	Placebo		
Kidney failure or ≥57% decline in eGFR	460	589	22·3	28·8	0·76 (0·68–0·86)	<0·0001
Kidney failure, ≥57% decline in eGFR, or cardiovascular death	612	763	29·7	37·3	0·79 (0·71–0·88)	<0·0001
Kidney failure, ≥57% decline in eGFR, heart failure hospitalisation, or cardiovascular death	786	997	38·6	49·6	0·77 (0·70–0·85)	<0·0001
Kidney failure	346	399	16·7	19·4	0·85 (0·74–0·99)	0·030
Dialysis or kidney transplantation	189	240	8·5	10·9	0·77 (0·64–0·94)	0·0089
Heart failure hospitalisation	263	334	12·0	15·3	0·78 (0·66–0·92)	0·0024
Heart failure hospitalisation or cardiovascular death	420	520	19·1	23·9	0·80 (0·70–0·91)	0·0006
Cardiovascular death	182	220	8·1	9·9	0·82 (0·67–0·999)	0·049
All-cause death	569	642	25·4	28·8	0·88 (0·79–0·99)	0·026

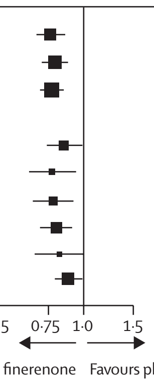


Figure 2: **Kidney, cardiovascular, and mortality outcomes in the overall INFINITY population**
Shown are prespecified kidney and cardiovascular efficacy outcomes, assessed in time-to-event analyses. Kidney failure was defined as sustained eGFR <15 mL/min per 1·73 m², initiation of chronic dialysis, or kidney transplantation. Cardiovascular death in this analysis excluded deaths classified as undetermined. eGFR=estimated glomerular filtration rate. HR=hazard ratio.

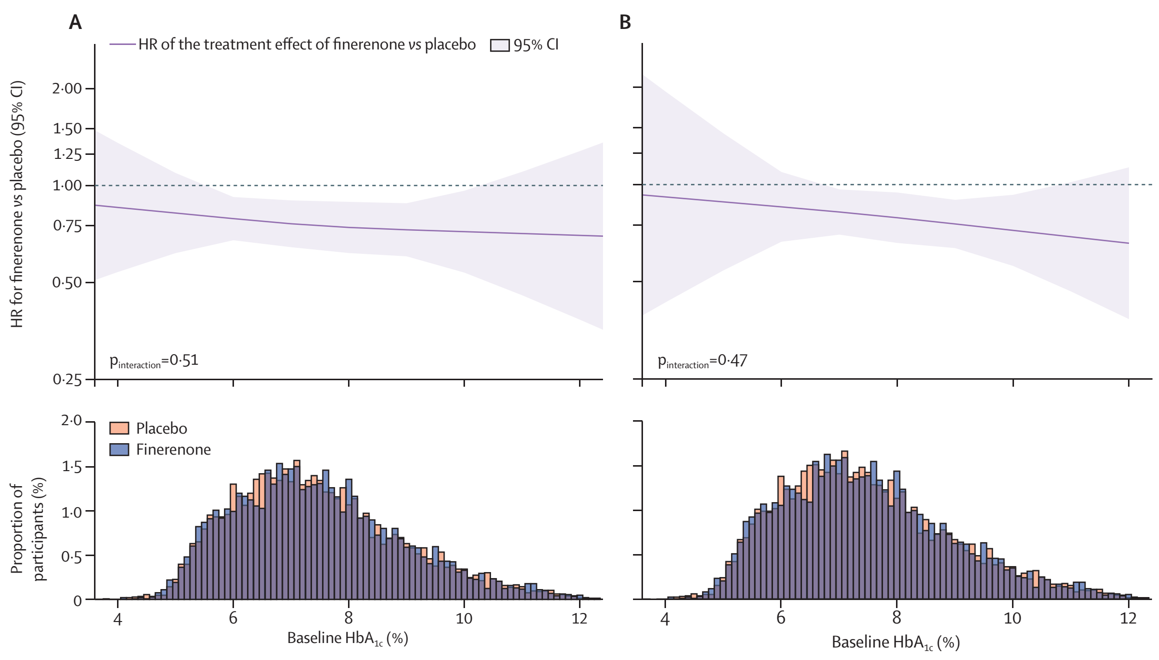


Figure 3: **Effects of finerenone on the composite kidney outcome and the composite cardiovascular outcome across the spectrum of glycated haemoglobin**
The solid line shows the HR for the treatment effect of finerenone relative to placebo on the main composite kidney outcome (A) and the main composite cardiovascular outcome (B), with the shading indicating the 95% CI. The kidney composite outcome included a sustained 57% or more decline in eGFR or kidney failure. The cardiovascular composite outcome included hospitalisation for heart failure or cardiovascular death. The distribution of baseline HbA_{1c} is shown in the histograms at the bottom of each panel. HbA_{1c}=glycated haemoglobin. HR=hazard ratio.

composite kidney outcome (≥57% decline in eGFR or kidney failure) and 745 (5·1%) who reached kidney failure. The incidence of the main composite kidney outcome was lower with finerenone than with placebo (22·3 *vs* 28·8 events per 1000 patient-years), corresponding to a hazard ratio (HR) of 0·76 (95% CI 0·68–0·86; p<0·0001; figure 1A). This effect was driven by all components of the composite kidney outcome (appendix p 22). This effect was similar when 40% or more and 50% or more declines in eGFR were substituted for a 57% or more decline in eGFR as part of the composite kidney outcome (appendix p 28). Finerenone also reduced the risk of kidney failure alone (0·85 [0·74–0·99]; p=0·030; figure 1B), and the post-hoc outcome comprising chronic dialysis or kidney transplantation (0·77 [0·64–0·94]; p=0·0089).

Finerenone reduced the risk of the main composite cardiovascular outcome (heart failure hospitalisation or cardiovascular death) versus placebo (19·1 *vs* 23·9 events per 1000 patient-years; HR 0·80 [95% CI 0·70–0·91]; p=0·0006; figure 1D). The risk of kidney and cardiovascular outcomes, including the composite kidney outcome inclusive of cardiovascular death, and inclusive of heart failure hospitalisation or cardiovascular death (ie, the main kidney and cardiovascular composite outcome), was lower with finerenone compared with placebo (figure 2). Finerenone also reduced the risk of heart failure hospitalisation alone (0·78 [0·66–0·92]; p=0·0024), cardiovascular death alone (0·82 [0·67–0·999]; p=0·049), and all-cause death (0·88 [0·79–0·99]; p=0·026; figure 2). Effects on other cardiovascular and hospitalisation outcomes, including non-fatal myocardial infarction, non-fatal stroke, heart failure hospitalisation, or cardiovascular death are in the appendix (p 29). Effects on key outcomes were consistent across trial populations (all interaction p≥0·40; appendix p 30).

The effects of finerenone on the main composite kidney and cardiovascular outcomes were consistent across the full spectrum of glycated haemoglobin ($p_{interaction}$=0·51 for kidney and $p_{interaction}$=0·47 for cardiovascular outcomes; figure 3). Likewise, the effect on the composite kidney outcome was consistent across a range of prespecified subgroups, including CKD aetiology, baseline eGFR, UACR, and SGLT2 inhibitor use (figure 4; appendix p 31). The effect on the composite cardiovascular outcome was similarly consistent across almost all prespecified subgroups (appendix p 32).

The incidence of adverse outcomes by treatment group is in table 2 and by diabetes status in the appendix (p 23). The incidence of serious adverse events, adverse events leading to permanent discontinuation, and acute kidney injury were similar with finerenone versus placebo. Investigator-reported hyperkalaemia occurred more

	Participants with an event/total participants: Finerenone	Participants with an event/total participants: Placebo	Rate per 1000 patient-years: Finerenone	Rate per 1000 patient-years: Placebo	HR (95% CI)	$p_{interaction}$
Diabetes status						**0·86**
Yes	356/6498	464/6492	19·5	25·6	0·76 (0·66–0·87)	
No	104/793	125/791	44·2	54·2	0·78 (0·60–1·01)	
Chronic kidney disease aetiology						**0·77**
Diabetes	356/6498	464/6492	19·5	25·6	0·76 (0·66–0·87)	
Glomerulonephritis	63/446	75/457	47·6	56·2	0·81 (0·58–1·14)	
Hypertension	24/234	36/225	34·3	55·4	0·62 (0·37–1·05)	
Unknown	14/84	13/85	57·7	51·1	1·00 (0·47–2·13)	
Other	3/29	1/24	..	..	..	
eGFR, mL/min per 1·73m²						**0·16**
<45	293/2617	349/2613	42·8	51·5	0·85 (0·73–0·99)	
45 to <60	71/1918	104/1931	13·2	19·3	0·67 (0·49–0·90)	
≥60	96/2755	136/2737	11·5	16·4	0·67 (0·51–0·87)	
UACR, mg/g						**0·24**
<300	40/2218	43/2155	5·6	6·3	0·88 (0·57–1·35)	
300–1000	93/2884	146/2917	11·6	17·9	0·63 (0·49–0·82)	
>1000	327/2187	400/2208	60·1	73·7	0·81 (0·70–0·93)	
SGLT2 inhibitor use						**0·99**
Yes	30/570	39/572	18·3	24·3	0·77 (0·48–1·23)	
No	430/6721	550/6711	22·7	29·2	0·77 (0·68–0·87)	
Overall	**460/7291**	**589/7283**	**22·3**	**28·8**	**0·76 (0·68–0·86)**	

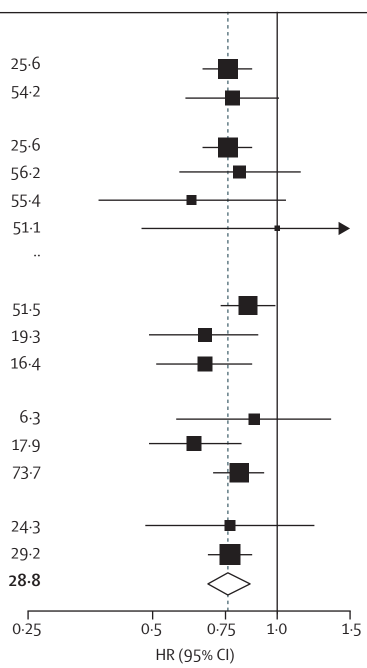


Figure 4: **Effect of finerenone on the composite kidney outcome across prespecified subgroups**
Analyses conducted based on stratified Cox models fitted across subgroup categories, including covariates treatment, subgroup, and the respective interaction. Heterogeneity between subgroups was assessed via the interaction p-value. eGFR=estimated glomerular filtration rate. HR=hazard ratio. SGLT2=sodium–glucose co-transporter-2. UACR=urinary albumin-to-creatinine ratio.

	Finerenone (n=7282)	Placebo (n=7265)
Any treatment-emergent adverse event	6124 (84·1%)	6109 (84·1%)
Any treatment-emergent adverse event leading to permanent discontinuation of study drug	431 (5·9%)	373 (5·1%)
Any treatment-emergent serious adverse event	2220 (30·5%)	2349 (32·3%)
Treatment-emergent serious adverse event leading to permanent discontinuation of study drug	146 (2·0%)	162 (2·2%)
Hypotension	350 (4·8%)	224 (3·1%)
Acute kidney injury*	243 (3·3%)	259 (3·6%)
Serious acute kidney injury requiring hospitalisation*	94 (1·3%)	94 (1·3%)
Any treatment-emergent hyperkalaemia	1043 (14·3%)	553 (7·6%)
Leading to permanent discontinuation of study drug	122 (1·7%)	39 (0·5%)
Any serious hyperkalaemia	77 (1·1%)	21 (0·3%)
Requiring hospitalisation	68 (0·9%)	15 (0·2%)
Life threatening	5 (0·1%)	5 (0·1%)
With fatal outcome	0	0
Any treatment-emergent hypokalaemia	74 (1·0%)	165 (2·3%)
Any serious hypokalaemia	3 (<0·1%)	11 (0·2%)
Requiring hospitalisation	2 (<0·1%)	11 (0·2%)
Potassium level		
<3·5 mmol/L	256/7058 (3·6%)	564/7035 (8·0%)
>5·5 mmol/L	1218/7154 (17·0%)	565/7128 (7·9%)
>6·0 mmol/L	245/7201 (3·4%)	98/7179 (1·4%)

Data are n (%) or n/N (%). Adverse events that started or worsened after the first dose of study drug up to 3 days after any temporary or permanent interruption of study drug were considered as treatment-emergent adverse events. *Preferred term.

Table 2: **Treatment-emergent adverse events**

frequently with finerenone (1043 [14·3%] of 7282) than with placebo (553 [7·6%] of 7265). However, hyperkalaemia leading to discontinuation of study treatment was uncommon (122 [1·7%] with finerenone *vs* 39 [0·5%] with placebo). Hyperkalaemia requiring hospitalisation occurred very infrequently (68 [0·9%] with finerenone *vs* 15 [0·2%] with placebo), and no fatal hyperkalaemia events were observed. The effect of finerenone on serious hyperkalaemia requiring hospitalisation was modified by diabetes status with increased risk observed in participants with diabetes versus without diabetes (HR 6·21 [95% CI 3·18–12·12], p<0·0001 *vs* 1·48 [0·47–4·67], p=0·50; $p_{interaction}$=0·034; appendix p 33). There was no imbalance in acute kidney injury between treatment groups and finerenone did not affect the risk of serious acute kidney injury requiring hospitalisation, with no interaction by diabetes status (1·01 [0·76–1·34], p=0·95; $p_{interaction}$=0·77; appendix p 33). Incidence of central laboratory-confirmed serum potassium by treatment group is in table 2.

Overall, for every 1000 patients treated for 3 years, finerenone was estimated to avert 34 kidney and cardiovascular composite outcomes, 21 kidney outcomes, 14 heart failure hospitalisations or deaths due to cardiovascular disease, and ten deaths from any cause. Because rates of heart failure hospitalisation and mortality were higher among participants with type 2 diabetes, absolute reductions in these outcomes were larger in these individuals compared with individuals with non-diabetic CKD. Irrespective of diabetes status, the numbers of composite kidney and cardiovascular outcomes averted with finerenone outweighed any absolute increases in hyperkalaemia (figure 5). The numbers needed to treat and numbers needed to harm for these outcomes are in the appendix (p 24).

Discussion

This prespecified pooled individual participant data analysis of three randomised, double-blind, placebo-controlled trials provides a comprehensive assessment of the efficacy and safety of finerenone in CKD. In the studied population, finerenone reduced the risk of CKD progression (including kidney failure alone) irrespective of glycaemic status, cause of kidney disease, eGFR, and albuminuria. Finerenone also reduced cardiovascular and mortality outcomes. As expected, the risk of hyperkalaemia was increased with finerenone; however, serious events leading to hospitalisation were uncommon. Although the competing risks of hospitalisation, mortality, and hyperkalaemia varied between participants with diabetic and non-diabetic CKD, the absolute benefits outweighed any harms across both populations. Taken together, these findings provide key evidence to support the role of finerenone as a foundational therapy for the prevention of kidney and cardiovascular events in patients with CKD.

Although trials of steroidal mineralocorticoid receptor antagonists have suggested potential reductions in albuminuria, results have been inconsistent, and studies to date have been too small or too short to provide reliable evidence on clinical outcomes or even eGFR slope.[1] In the only completed large trial of a steroidal mineralocorticoid receptor antagonist in 1434 patients with CKD, spironolactone did not reduce cardiovascular events or slow eGFR decline.[16] Notably, two thirds of participants assigned to spironolactone discontinued treatment within 6 months based on protocolised safety criteria, primarily related to worsening kidney function or hyperkalaemia.[16] These findings underscore the need for more effective and safer approaches to blocking the deleterious effects of excess aldosterone in patients with CKD.

The KDIGO guidelines recommend finerenone for patients with CKD related to type 2 diabetes and persistent albuminuria despite first-line treatment with a renin–angiotensin system inhibitor and an SGLT2 inhibitor.[8] The consistent effects of finerenone on kidney and cardiovascular outcomes regardless of glycaemic status, CKD aetiology, eGFR, and albuminuria support extension of its role to a substantially broader CKD population. Importantly, although participants without diabetes comprised a relatively modest proportion of the overall pooled population, the 225 composite kidney events contributed by this group accounted for one-fifth of all kidney outcomes, reinforcing the robustness of the treatment effect estimates for this outcome. The

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consistent reduction in CKD progression regardless of SGLT2 inhibitor use supports the value of combination therapy with both agents to reduce kidney and cardiovascular risk in this population. In addition, the clear and separate reduction in kidney failure with finerenone (including initiation of dialysis or transplantation) is notable, because avoidance of dialysis or kidney transplantation is a universally important outcome for patients with CKD and clinicians alike. Taken together, these data provide a compelling rationale for updating nephrology guideline recommendations to extend the use of finerenone to a broader range of CKD aetiologies.

Reductions in heart failure and mortality outcomes in this analysis are complemented by data from the FINEARTS-HF trial, which showed that finerenone reduces the risk of heart failure events among patients with heart failure with mildly reduced or preserved ejection fraction, irrespective of diabetes status.[17,18] This finding is important because heart failure with preserved ejection fraction represents the dominant phenotype of heart failure in CKD, becoming increasingly prevalent as kidney function declines.[19] Moreover, most patients with CKD die from cardiovascular causes before reaching kidney failure.[20] Although the population studied in FINEARTS-HF was overall at low risk of kidney outcomes, in participants with severely increased albuminuria (UACR ≥300 mg/g), finerenone attenuated the chronic rate of eGFR decline.[21] Collectively these data underscore the potential for finerenone to improve multiple kidney and cardiovascular outcomes across overlapping populations at high risk.

The pooled INFINITY population highlights the markedly different absolute risks of key outcomes in patients with CKD depending on the presence or absence of type 2 diabetes. The substantially higher risk of cardiovascular events in individuals with type 2 diabetes, particularly for heart failure, supports prioritising rapid, or even simultaneous, initiation of finerenone alongside other guideline-directed therapies such as SGLT2 inhibitors. Pooled analyses from FIDELIO-DKD and FIGARO-DKD demonstrate that reductions in heart failure events are observed within the first 6 months of treatment.[22] The CONFIDENCE trial further showed that simultaneous initiation of finerenone and empagliflozin produced greater reductions in albuminuria than either therapy alone, without increasing the risk of acute kidney injury.[23] Although patients with type 2 diabetes have a higher risk of hyperkalaemia, driven in part by type 4 renal tubular acidosis,[24] the absolute risk of hyperkalaemia leading to hospitalisation was low with no deaths attributable to hyperkalaemia, providing reassurance regarding the safety of implementing finerenone in routine practice.

Although these data represent the totality of the high-quality randomised evidence for finerenone in CKD, there are some important limitations to recognise.

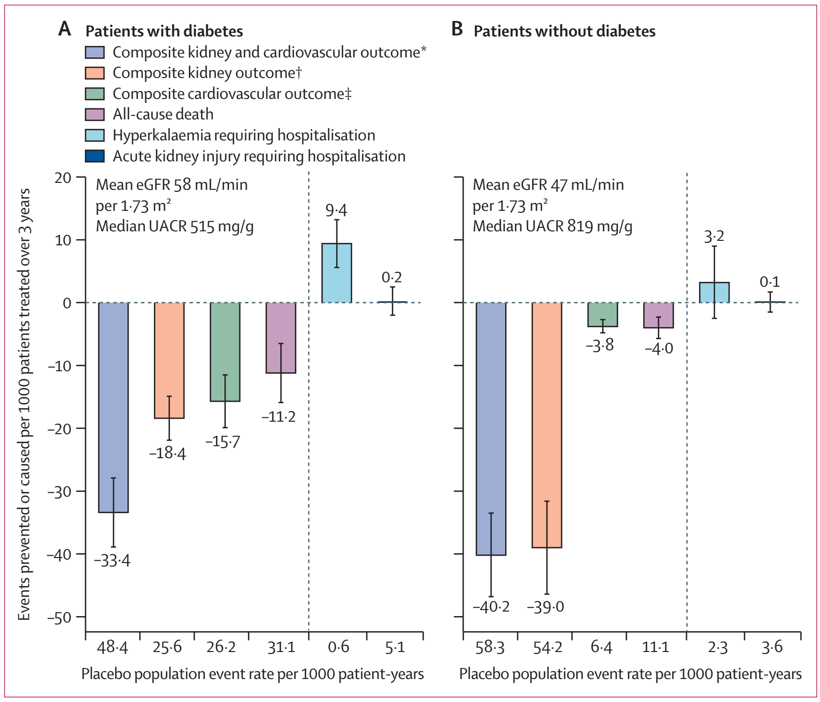

***Figure 5:* Estimated absolute effects of finerenone per 1000 patients treated over 3 years by diabetes status**
Estimated absolute numbers of events prevented (negative numbers) or caused (positive values) per 1000 patients treated for 3 years are shown. Error bars represent SE estimated from uncertainty in the hazard ratios. Placebo population event rates are the numbers of events per 1000 patient-years in the placebo groups of all trials in the relevant subpopulation. eGFR=estimated glomerular filtration rate. UACR=urinary albumin-to-creatinine ratio. *The composite kidney outcome, or heart failure hospitalisation or cardiovascular death. †Sustained 57% or more decline in eGFR relative to baseline or kidney failure. Kidney failure was defined as sustained eGFR <15 mL/min per 1·73 m² or initiation of chronic dialysis or kidney transplantation. ‡Heart failure hospitalisation or cardiovascular death. Cardiovascular death in this analysis excluded deaths classified as undetermined.

Patients with non-albuminuric CKD were not studied in the three included trials and thus effects on CKD progression in this population are unknown. Although 903 patients with glomerular diseases were included in FIND-CKD, patients with lupus nephritis and anti-neutrophil cytoplasmic antibody-associated vasculitis were excluded, as were individuals requiring immunosuppression and individuals with cystic kidney diseases. Although patients with type 1 diabetes were excluded, a separate trial in this population showed that finerenone reduced albuminuria, indicating a potential role in preserving kidney function in patients with CKD due to type 1 diabetes.[25] Although effects on cardiovascular outcomes were consistent across a range of prespecified subgroups, there were relatively few events in patients without diabetes and thus overall effect estimates for these outcomes were largely driven by populations with type 2 diabetes. Although subgroup analyses were prespecified, tests for interaction were not corrected for multiple comparisons, increasing the possibility of chance findings. Because these trials were commenced before SGLT2 inhibitors were guideline-recommended in CKD,[5,6,13,26,27] there was a relatively small proportion of individuals receiving SGLT2 inhibitors at baseline; however, the consistency of effects by baseline SGLT2 inhibitor use is entirely in keeping with data for finerenone in heart failure.[28] Although cardiovascular and mortality outcomes were not adjudicated in FIND-CKD, events were prospectively collected using structured electronic case report forms incorporating standardised cardiovascular event definitions, consistent with regulatory recommendations for cardiovascular outcome assessment. Absolute effect estimates reflect baseline risk in the studied populations and are not generalisable to other CKD populations. Finally, the trials did not enrol patients with eGFR of less than 25 mL/min per 1·73 m² at screening and thus the efficacy and safety of initiating finerenone below this threshold is unknown.

In summary, in the studied CKD populations, finerenone reduced the risk of CKD progression, including kidney failure alone, and reduced cardiovascular events and mortality across a broad range of disease aetiologies and levels of glycaemia, eGFR, and albuminuria.

Contributors

BLN and RA wrote the first draft of the report with input from all authors. JDS and NJ did the statistical analysis, independently confirmed by CA and PSc. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication. JDS, NJ, HJLH, CA, and PSc directly accessed and verified the underlying data reported in the manuscript.

Declaration of interests

BLN has received grants for publication support from Menarini (paid directly to George Institute of Global Health); has received consultancy fees from AstraZeneca, Alexion, Bayer, Boehringer and Ingelheim, CSL-Seqirus, Novo Nordisk, Proton Intelligence, Sobi, Travere, Otsuka, and Vera Therapeutics; has received payment or honoraria from AstraZeneca, Boehringer Ingelheim, Bayer, MJH Life Sciences, and Cornerstone Medical Education; has received support for meeting attendance or travel from AstraZeneca, CSL Behring, Novo Nordisk; and is the Deputy Scientific Chair, World Congress of Nephrology 2026. HJLH has received honoraria from Bayer for participation in the FINE-ONE clinical trial steering committee (honoraria paid to employer University of Groningen); received grants from AstraZeneca, Bayer, Boehringer Ingelheim, Janssen, and Novo Nordisk (paid to employer University of Groningen); received consulting fees from AstraZeneca Alexion, Alnylam, Amgen, Bayer, Boehringer Ingelheim, Biocity Biopharmaceuticals, Dimerix, Eli Lilly, Novo Nordisk, Novartis, Roche, and Travere Therapeutics; received speaker fees from AstraZeneca, Bayer, and Novo Nordisk; and received support for attending meetings or travel from AstraZeneca and Eli Lily. VP has received honoraria for steering committee, data monitoring committee, or advisory board roles or for scientific presentations from AstraZeneca, Bayer, Biogen, Boehringer Ingelheim, Cajal Consulting, Emerald Clinical, GlaxoSmithKline, Guard therapeutics, Incyte, Janssen, Mineralys, Novo Nordisk, Novartis, Otsuka, Purespring, Sidera, Sitala, Shanxi Micot, Travere, Tricida, and Vifor; serves on the board of directors for Emerald Clinical, St Vincents Health Australia, Kidney Health Australia, and Mindgardens Neuroscience Network; and has stock in Emerald Clinical. DZIC has received research grants from AstraZeneca, Bayer, Boehringer Ingelheim/Eli Lilly, CSL Behring, Janssen, Merck, Novo Nordisk, and Sanofi; has received consultancy fees from Boehringer Ingelheim/Eli Lilly, Merck, AstraZeneca, Sanofi, Mitsubishi-Tanabe, AbbVie, Janssen, Amgen, Bayer, Prometic, Bristol Myers Squibb, Maze, Gilead, CSL-Behring, Otsuka, Novartis, Youngene, Lexicon, Inversago, GlaxoSmithKline, Biobridge, Vantage, Altimmune, and Novo Nordisk; has received payment honoraria from AstraZeneca, Bayer, Boehringer Ingelheim/Eli Lilly, Janssen, Merck, Mitsubishi Tanabe, Novo Nordisk, and Sanofi; has received meeting travel or attendance support from AstraZeneca, Bayer, Boehringer Ingelheim/Eli Lily, Janssen, Merck, and Novo Nordisk; and was in receipt of drugs from AstraZeneca. 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CW has received consultancy fees from Vera Therapeutics and Bayer; has received honoraria from Boehringer Ingelheim and Eli Lilly; participated in Data Safety Monitoring Board or Advisory Board for AstraZeneca, Bayer, GlaxoSmithKline, Novo Nordisk, Sanofi, CSL-Vifor, and Merck. PSa has received honoraria for advisory boards from Astellas, Bayer, Boehringer Ingelheim, Value Research, Vantive, TEVA, Specialty Therapeutics, PRO.MED.CS, F Hoffmann-La Roche, and SOBI; speaker fees from AstraZeneca, Bayer, Genesis Pharma, Boehringer Ingelheim, Winmedica, Vifor Fresenius, and Vantive; research support from AstraZeneca, Boehringer Ingelheim, Genesis Pharma, Vianex, and Menarini; has received meeting travel or attendance support from Bayer, Specialty Therapeutics, and Genesis Pharma; and is a member of trial committees for Bayer, CSL Behring, and Alnylam/F Hoffmann-La Roche. SDA has received research support from Abbott Vascular and Vifor Pharma; has received consultancy fees from AstraZeneca, Berlin Heals, BiMyo, Brahms, Cordio, Corvia, Aytokinetics, Edwards, Novartis, Novo Nordisk, Regeneron, Relaxera, Scirent, Sensible Medical, Vectorious, Veru, and Viscardia; has received honoraria from Actimed Therapeutics, Eli Lily, Mankind Pharma, Reparion, and Vivus; is named co-inventor of two patent applications regarding MR-proANP (DE 102007010834 & DE 102007022367), but does not benefit personally from the related issued patents; and has a leadership or fiduciary role for Alleviant, Amgen, Bayer AG, Boehringer Ingelheim, Cardiac Dimensions, Cardior, CSL-Vifor, CVRx, Impluse Dynamics, Occlutech, Pfizer, Pulnovo and V-Wave. GF reports grants from the European Commission; honoraria from Bayer, Boehringer Ingelheim, Servier, and Novo Nordisk; participates in Data Safety Monitoring or Advisory Board for Bayer, Boehringer Ingelheim, and Novo Nordisk; acts in a leadership or fiduciary role for Heart Failure Association, JACC-Heart Failure, European Journal of Heart Failure; and receives trial committee membership fees from Medtronic, Bayer, Boehringer Ingelheim, Vifor, Amgen, Servier, Impulse Dynamics, Cardior, Merck, AstraZeneca, and Novo Nordisk. BP reports consultant fees from Bayer, Boehringer Ingelheim, Lexicon, AstraZeneca, Vifor, ScPharmaceuticals, SQ innovations, G3 Pharmaceuticals, Sarfez Pharmaceuticals, KBP Biosciences, Cereno Scientific, Brainstorm Medical, Prointel, PhaseBio, Tricida, and Sea Star Medica; has stock options with G3 Pharmaceuticals, Brainstorm Medical, Proton Intelligence, Tricida, Vifor, ScPharmaceuticals, SQ innovations, Sarfez Phaemaceuticals, KBP Biosciences, and Cereno Scientific; holds a patent for site-specific delivery of eplerenone to the myocardium (US patent 9931412) and a provisional patent for histone acetylation-modulating agents for the treatment and prevention of organ injury (provisional patent US 63/045,784). PR reports sponsorship from Bayer during the conduct of the study; has received research grants from AstraZeneca, Bayer, and Novo Nordisk; has received consultancy fees from Astellas Pharma, Abbott, Amgen, Boehringer Ingelheim, Eli Lilly, Gilead, Mundipharma, Novartis, Roche, Regeneron, Sanofi, and Vifor Pharma; received honoraria from Bayer, Eli Lilly, and Daiichi Sankyo; has received meeting travel or attendance support from Bayer; received study medication from Bayer, AstraZeneca, Lexicon Pharma, and Novo Nordisk; and has financial or non-financial interests with Astellas Pharma, Gilead, Mundipharma, Novartis, Regeneron, Sanofi, and Vifor Pharma; all fees are given to Steno Diabetes Center Copenhagen. LMR reports consultant fees from Bayer. NJ received travel support from AstraZeneca. JDS has no competing interests. MB, CA, AH-B, PSc, TE, RD, and RL are employees of Bayer, Germany, and report ownership interest in Bayer. AL is a full-time employee at Bayer, Brazil, and reports ownership interest in Bayer. RA has received support from Bayer for Steering committee, adjudication committee, publication committee of FIDELIO-DKD and FIGARO-DKD, and being a Principal Investigator for FIDELITY analysis; has had ad hoc consultant, speaker, and participation in Advisory Boards; has received consultancy fees from Boehringer Ingelheim, Novartis, Akebia, Intercept Pharma, and Alnylam; has received meeting travel or attendance support from Boehringer Ingelheim, Novartis, Akebia, and Vertex; and participates on a Drug Safety Monitoring or Advisory Board for Vertex, Eloxx, and Chinook.

Data sharing

Availability of the data underlying this publication will be determined according to Bayer's commitment to the EFPIA–PhRMA principles for responsible clinical trial data sharing, which pertains to scope, timepoint, and process of data access. As such, Bayer commits to sharing upon request from qualified scientific and medical researchers patient-level clinical trial data, study-level clinical trial data, and protocols from clinical trials in patients for medicines and indications approved in the US and EU as necessary for conducting legitimate research. This commitment applies to data on new medicines and indications that have been approved by the EU and US regulatory agencies on or after Jan 01, 2014. Interested researchers can request access at https://vivli.org/ to anonymised patient-level data and supporting documents from clinical studies to conduct further research that can help advance medical science or improve patient care. Information on the Bayer criteria for listing studies and other relevant information is provided in the member section of the portal. Data access will be granted to anonymised patient-level data, protocols and clinical study reports after approval by an independent scientific review panel. Bayer is not involved in the decisions made by the independent review panel. Bayer will take all necessary measures to ensure that patient privacy is safeguarded.

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